National Testing Agency

Question Paper Name :	B Tech 2nd Apr 2026 Shift 2
Subject Name :	B. Tech
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Duration :	180
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Display Marks:	Yes

B. Tech

Group Number :	1
Group Id :	6911213
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Id :	69112113
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112113
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 691121151 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let α, β be the roots of the equation $x^2 - 3x + r = 0$, and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 + 3x + r = 0$.

If the roots of the equation $x^2 + 6x = m$ are $2\alpha + \beta + 2r$ and $\alpha - 2\beta - \frac{r}{2}$, then m is equal to :

Options :

691121511. -135

691121512. -567

691121513. 135

691121514. 567

Question Number : 2 Question Id : 691121152 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the circles $C_1: |z| = r$ and $C_2: |z-3-4i| = 5$, $z \in \mathbb{C}$, be such that C_2 lies within C_1. If z_1 moves on C_1, z_2 moves on C_2 and $\min|z_1 - z_2| = 2$, then $\max|z_1 - z_2|$ is equal to :

Options :

691121515. 12

691121516. 17

691121517. 22

691121518. 24

Question Number : 3 Question Id : 691121153 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the system of equations

$x + 5y + 6z = 4,$

$2x + 3y + 4z = 7,$

$x + 6y + az = b$

has infinitely many solutions, then the point (a, b) lies on the line

Options :

691121519. $y - x = 3$

691121520. $x - y = 3$

691121521. $x + y = 11$

691121522. $x + y = 12$

Question Number : 4 Question Id : 691121154 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $a_1, a_2, a_3, \ldots$ be an A.P. and $g_1 = a_1, g_2, g_3, \ldots$ be an increasing G.P. If $a_1 = a_2 + g_2 = 1$ and $a_3 + g_3 = 4$, then $a_{10} + g_5$ is equal to :

Options :

691121523. 81

691121524. 76

691121525. 62

691121526. 55

Question Number : 5 Question Id : 691121155 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The sum $\frac{1^3}{1}+\frac{1^3+2^3}{1+3}+\frac{1^3+2^3+3^3}{1+3+5}+\cdots$ up to 8 terms, is :

Options :

691121527. 70

691121528. 71

691121529. 72

691121530. 73

Question Number : 6 Question Id : 691121156 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If for $3 \leq r \leq 30$, $\left({}^{30}C_{30-r}\right)+3\left({}^{30}C_{31-r}\right)+3\left({}^{30}C_{32-r}\right)+\left({}^{30}C_{33-r}\right)={}^{m}C_{r}$, then m equals :

Options :

691121531. 31

691121532. 32

691121533. 33

691121534. 34

Question Number : 7 Question Id : 691121157 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let p_n denote the total number of triangles formed by joining the vertices of an n-side regular polygon. If $p_{n+1}-p_n=66$, then the sum of all distinct prime divisors of n is :

Options :

691121535. 7

691121536. 8

691121537. 5

691121538. 6

Question Number : 8 Question Id : 691121158 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A man throws a fair coin repeatedly. He gets 10 points for each head he throws and 5 points for each tail he throws. If the probability that he gets exactly 30 points is $\frac{m}{n}$, gcd $(m, n)=1$, then $m+n$ is equal to :

Options :

691121539. 53

691121540. 55

691121541. 107

691121542. 105

Question Number : 9 Question Id : 691121159 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The mean and variance of n observations are 8 and 16, respectively. If the sum of the first $(n-1)$ observations is 48 and the sum of squares of the first $(n-1)$ observations is 496, then the value of n is :

Options :

691121543. 21

691121544. 16

691121545. 13

691121546. 7

Question Number : 10 Question Id : 691121160 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let a circle pass through the origin and its centre be the point of intersection of two mutually perpendicular lines $x+(k-1)y+3=0$ and $2x+k^2y-4=0$. If the line $x-y+2=0$ intersects the circle at the points A and B, then $(AB)^2$ is equal to :

Options :

691121547. 10

691121548. 27

691121549. 18

691121550. 34

Question Number : 11 Question Id : 691121161 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let O be the origin, and P and Q be two points on the rectangular hyperbola $xy = 12$ such that the mid point of the line segment PQ is $\left(\frac{1}{2}, -\frac{1}{2}\right)$. Then the area of the triangle OPQ equals :

Options :

691121551. $\frac{3}{2}$

691121552. $\frac{5}{2}$

691121553. $\frac{7}{2}$

691121554. $\frac{9}{2}$

Question Number : 12 Question Id : 691121162 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the parabola $y = x^2 + px + q$ passing through the point $(1, -1)$ be such that the distance between its vertex and the x-axis is minimum. Then the value of $p^2 + q^2$ is :

Options :

691121555. 2

691121556. 4

691121557. 5

691121558. 8

Question Number : 13 Question Id : 691121163 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $P = \{\theta \in [0, 4\pi] : \tan^2\theta \neq 1\}$ and $S = \{a \in \mathbf{Z} : 2(\cos^8\theta - \sin^8\theta)\sec 2\theta = a^2, \theta \in P\}$. Then n(S) is :

Options :

691121559. 0

691121560. 1

691121561. 2

691121562. 3

Question Number : 14 Question Id : 691121164 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the vectors $\vec{a} = -\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$. For some $\lambda, \mu \in \mathbf{R}$, let $\vec{c} = \lambda\vec{a} + \mu\vec{b}$.

If $\vec{c} \cdot \left(3\hat{i} - 6\hat{j} + 2\hat{k}\right) = 10$ and $\vec{c} \cdot \left(\hat{i} + \hat{j} + \hat{k}\right) = -2$, then $|\vec{c}|^2$ is equal to :

Options :

691121563. 8

691121564. 12

691121565. 14

691121566. 15

Question Number : 15 Question Id : 691121165 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the point A be the foot of perpendicular drawn from the point P(a, b, 0) on the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-\alpha}{3}$. If the midpoint of the line segment PA is $\left(0, \frac{3}{4}, \frac{-1}{4}\right)$, then the value of $a^2+b^2+\alpha^2$ is equal to :

Options :

691121567. 1

691121568. 2

691121569. 6

691121570. 9

Question Number : 16 Question Id : 691121166 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two adjacent sides of a parallelogram PQRS are given by $\vec{PQ} = \hat{j} + \hat{k}$ and $\vec{PS} = \hat{i} - \hat{j}$. If the side PS is rotated about the point P by an acute angle α in the plane of the parallelogram so that it becomes perpendicular to the side PQ, then $\sin^2\left(\frac{5\alpha}{2}\right) - \sin^2\left(\frac{\alpha}{2}\right)$ is equal to :

Options :

691121571. $\frac{1}{2}$

691121572. $\frac{\sqrt{3}}{2}$

691121573. $\frac{\sqrt{3}}{4}$

691121574. $\frac{2\sqrt{3}}{5}$

Question Number : 17 Question Id : 691121167 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The value of $\int_0^{20\pi}(\sin^4 x + \cos^4 x)dx$ is equal to :

Options :

691121575. $\frac{15\pi}{2}$

691121576. 25π

691121577. 15π

691121578. $\frac{25\pi}{2}$

Question Number : 18 Question Id : 691121168 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f(x)$ be a polynomial of degree 5, and have extrema at $x=1$ and $x=-1$. If $\lim\limits_{x\to 0}\left(\frac{f(x)}{x^3}\right)=-5$, then $f(2)-f(-2)$ is equal to :

Options :

691121579. 0

691121580. 50

691121581. 92

691121582. 112

Question Number : 19 Question Id : 691121169 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f(x)=\int\left(\frac{16x+24}{x^2+2x-15}\right)dx$. If $f(4)=14\log_e(3)$ and $f(7)=\log_e(2^{\alpha}\cdot 3^{\beta})$, $\alpha, \beta \in \mathbf{N}$, then $\alpha+\beta$ is equal to :

Options :

691121583. 31

691121584. 37

691121585. 39

691121586. 41

Question Number : 20 Question Id : 691121170 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $x = x(y)$ be the solution of the differential equation $2y^2\frac{dx}{dy} - 2xy + x^2 = 0,\ y > 1, x(e) = e$.

Then $x(e^2)$ is equal to :

Options :

691121587. $\frac{3}{2}e^2$

691121588. $\frac{2}{3}e^2$

691121589. e^2

691121590. $2e^2$

Mathematics Section B

Section Id :	69112114
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112114
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Id : 691121171 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $A = \{2, 3, 4, 5, 6\}$. Let R be a relation on the set $A \times A$ given by $(x, y)R(z, w)$ if and only if x divides z and $y \leq w$. Then the number of elements in R is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Question Number : 22 Question Id : 691121172 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the matrices $A=\begin{bmatrix}2 & -2\\ 4 & -2\end{bmatrix}$ and $B=\begin{bmatrix}3 & 9\\ 1 & 3\end{bmatrix}$. If matrices P and Q are such that $PA=B$ and $AQ=B$, then the absolute value of the sum of the diagonal elements of $2(P+Q)$ is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 23 Question Id : 691121173 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let A be the point (3, 0) and circles with variable diameter AB touch the circle $x^2+y^2=36$ internally. Let the curve C be the locus of the point B. If the eccentricity of C is e, then $72e^2$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 24 Question Id : 691121174 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If the area of the region bounded by $16x^2-9y^2=144$ and $8x-3y=24$ is A, then $3(A+6\log_e(3))$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 25 Question Id : 691121175 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The number of points in the interval [2, 4], at which the function $f(x)=\left[x^2-x-\frac{1}{2}\right]$, where $[\cdot]$ denotes the greatest integer function, is discontinuous, is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Physics Section A

Section Id : 69112115